



Republic of the Philippines
OFFICE OF THE PRESIDENT
COMMISSION ON HIGHER EDUCATION
HEDC Bldg., C. P. Garcia Ave., UP Diliman, Q.C.

CERTIFICATE OF PROGRAM COMPLIANCE (COPC)

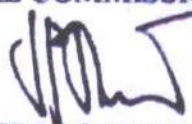
No.: 017
Series of 2019

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "*Higher Education Act of 1994*," and by virtue of Resolution No. 405-2019, of the Commission en banc, this Certificate of Program Compliance (COPC) certifies that **WESTERN MINDANAO STATE UNIVERSITY**, Zamboanga City, complies with the prescribed minimum requirements of CHED to offer the **MASTER IN INFORMATION TECHNOLOGY** program effective Academic Year 2019-2020.

This Certificate of Program Compliance (COPC), however, is subject to revocation if the herein institution fails to maintain the prescribed standards for the program. Furthermore, this COPC does not extend to any branch or satellite campus of the institution.

Quezon City, Philippines, August 28, 2019

FOR THE COMMISSION:


J. PROSPERO E. DE VERA III, DPA
Chairman

(NOT VALID WITHOUT SEAL)





A. PROGRAM

A.1 Program Design

Program Description

Master in Information Technology emphasizes the acquisition of concepts and technologies preparing and enable the student for the industrial practice of systems integration, systems administration, systems planning, systems implementation, and other activities that maintain the integrity and proper functionality of a system and its components.

Program Curriculum

SUMMARY OF SUBJECT COMPONENTS	UNITS
IT CORE COURSES (12 units)	
• MIT 201 – Advanced Operating System and Networking	3
• MIT 202 – Advanced Systems Design and Implementation	3
• MIT 203 – Advanced Database Systems	3
• MIT 204 – Technology and Project Management	3
	12
IT SPECIALIZATION COURSES (21 units)	
• MIT 205 – Technical Research	3
• MIT 206 – Statistical Research	3
• MIT 207 – Specialization Course 1 (System Development)	3
○ E-Commerce Solutions	
○ E-Learning Solutions	
• MIT 208 – Specialization Course 2 (Information Management [IM])	3
○ IM Modelling and Design	
○ IM Implementation, Configuration, and Administration	
• MIT 209 – Specialization Course 3 (Interactive Systems)	3
○ Multimedia Solutions	
○ Human-Computer Interaction	
• MIT 210 – Specialization Course 4 (Intelligent Systems)	3
○ Expert Systems	
○ Artificial Intelligence	
• MIT 211 – Specialization Course 5 (Emerging Trends in Computing)	3
	21
CAPSTONE PROJECT (6 units)	
• MIT 212A – Capstone Project 1- <i>Directed research</i>	3
• MIT 212B – Capstone Project 2 – <i>to be presented in a national or international public forum</i>	3
	6
TOTAL NUMBER OF UNITS	39



MASTER IN INFORMATION TECHNOLOGY

Effective S.Y. 2019–2020

PROGRAM OF STUDY		
SUBJECT CODE	DESCRIPTIVE TITLE	UNITS
FIRST YEAR – First Semester		
MIT 201	Advanced Operating System and Networking	3
MIT 202	Advanced System Design and Implementation	3
MIT 203	Advanced Database Systems	3
MIT 204	Technology and Project Management	3
		12
FIRST YEAR – Second Semester		
MIT 205	Technical Research	3
MIT 206	Statistical Research	3
MIT 207	Specialization Course 1	3
MIT 208	Specialization Course 2	3
		12
SECOND YEAR – First Semester		
MIT 209	Specialization Course 3	3
MIT 210	Specialization Course 4	3
MIT 211	Specialization Course 5	3
MIT 212A	Capstone Project 1	3
		12
SECOND YEAR – Second Semester		
MIT 212B	Capstone Project 2	3
		3
TOTAL NUMBER OF UNITS		39



Republic of the Philippines
Western Mindanao State University
COLLEGE OF COMPUTING STUDIES
GRADUATE PROGRAM – MASTER IN INFORMATION TECHNOLOGY



Course Description

COURSE CODE:	MIT201
COURSE TITLE:	Advanced Operating System and Networking
COURSE DESCRIPTION:	This course provides operating system concepts, process scheduling, CPU scheduling, deadlock, memory management, and file systems. Students will study widely used operating systems such as Windows and UNIX and other types of operating systems. Topics include kernel design, the I/O system, scheduling algorithms, process control, inter-process communication, system calls, memory management, and protection and security issues. Also, this course covers the different network protocols, setting and managing network connectivity, and network topologies.

COURSE CODE:	MIT202
COURSE TITLE:	Advanced Systems Design and Implementation
COURSE DESCRIPTION:	This course focuses on the analysis, logical design, and implementation of computer-based information systems. The emphasis is placed upon the development of requirements specifications that serve the business needs of the organization and provide the necessary base for subsequent systems development. The course also provides advanced topics particularly on the design using an object-oriented approach bridging from analysis to implementation. At the end of the course, students will learn how to effectively analyze, design, and implement information systems based on user requirements and needs.

COURSE CODE:	MIT203
COURSE TITLE:	Advanced Database Systems
COURSE DESCRIPTION:	This course provides a solid foundation and advances in database technology and its implementation. The course covers fundamental concepts of database technology such as the different approaches to database application development, conceptual and logical data models, physical database design, relational algebra, functional dependencies, constraints and triggers, NoSQL, Parallel and Distributed Databases, Transactions, and SQL Query Optimization.



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COURSE CODE:	MIT204
COURSE TITLE:	Technology and Project Management
COURSE DESCRIPTION:	The course covers the fundamental project management principles and methodologies for managing the software development life cycle and process models. It will also cover topics on process metrics, software project planning, monitoring, control, scheduling mechanisms, budget estimates, risk assessment, leadership, motivation, and teaching building.

COURSE CODE:	MIT205
COURSE TITLE:	Technical Research
COURSE DESCRIPTION:	This course introduces and discusses the fundamental concepts and methods of research particularly its meaning, types, methodology, review of literature and studies, analysis and interpretation of data, presentation of data, and writing conclusions and recommendations. In this course, the students will consider how to select the appropriate methodology to use in a chosen study. Additionally, these students will learn how to collect data based on different data collection methods, construct these tools, and pilot them before they become ready for use. Finally, this course elucidates the requirements for academic work, considering aspects related to language, writing style, and layout. To culminate this final stage, students will learn to write a comprehensive research proposal that may be conducted in the future

COURSE CODE:	MIT206
COURSE TITLE:	Statistical Research
COURSE DESCRIPTION:	The course deals with the basic principles of statistics, the presentation and analysis of data, and descriptive and inferential statistics that are necessary for the conduct of quantitative research. The subject also focuses on providing the students with the proper foundation to propose and conduct theoretical and empirical research to answer a broad array of questions related to theories and concepts of information and communications technology.



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COURSE CODE:	MIT207
COURSE TITLE:	Specialization Course 1 (System Development)
COURSE DESCRIPTION:	This course deals with the advanced study of methods and best practices in system development. It involves an in-depth analysis of different issues on organizational, business, and socio-technical issues in the implementation of IT, and how to effectively manage these issues related to theories and concepts of information technology.

COURSE CODE:	MIT208
COURSE TITLE:	Specialization Course 2 (Information Management)
COURSE DESCRIPTION:	This course focuses on the advancement of information management processes and their design and analysis using relevant data management technologies to efficiently perform the acquisition, organization, interpretation, and visualization of data among business industries. This also covers advances in data and process modeling techniques, database design, and query optimizations. Lastly, enables students to implement well-planned, IT-based systems that can be used for medium to large-scale business transactions.

COURSE CODE:	MIT209
COURSE TITLE:	Specialization Course 3 (Interactive Systems)
COURSE DESCRIPTION:	Interactive systems are computer systems characterized by significant amounts of interaction between humans and computers. This course concentrates solely on theories, principles, and methodologies of interactive system design and evaluation. Important topics to be covered include user-centered design, experience design, usability analysis, evaluation, and task analysis. With this, students will gain the knowledge, skills, and techniques necessary to design innovative interactive products to address the needs and accessibility of the target population to make the products more useful and usable for them.



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COURSE CODE:	MIT210
COURSE TITLE:	Specialization Course 4 (Intelligent Systems)
COURSE DESCRIPTION:	This course focuses on the advancement of information management processes and their design and analysis using relevant data management technologies to efficiently perform the acquisition, organization, interpretation, and visualization of data among business industries. This also covers advances in data and process modeling techniques, database design, and query optimizations. Lastly, enables students to implement well-planned, IT-based systems that can be used for medium to large-scale business transactions.

COURSE CODE:	MIT211
COURSE TITLE:	Specialization Course 5 (Emerging Trends in Computing)
COURSE DESCRIPTION:	This course introduces modern technologies used in the industry to solve complex and challenging problems. It involves discussions of modern algorithms based on Machine Learning and Artificial Intelligence, and their underlying concepts. At the end of the course, students will be able to apply these concepts to real-world problems to provide meaningful solutions.

COURSE CODE:	MIT212A
COURSE TITLE:	Capstone Project 1
COURSE DESCRIPTION:	Capstone Project focuses on the infrastructure application or processes involved in implementing a Computing solution to a problem. This course requires students to propose a capstone project that will introduce either a new application, technology, or process that can be used as a tool to solve an organizational or societal problem.

COURSE CODE:	MIT212B
COURSE TITLE:	Capstone Project 2
COURSE DESCRIPTION:	This course requires students to present the final capstone project that includes the system and documentation. The capstone project should be able to produce innovative results, generate new knowledge or theories or explore new frontiers of knowledge or application areas.